



PAK-AUSTRIA FACHHOCHSCHULE
INSTITUTE OF APPLIED SCIENCES AND TECHNOLOGY

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SUBJECT: Digital Image Processing

SECTION: AI (Green)

DEPARTMENT: IT & CS

LAB NO: 01

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Task 1: Basic Conda Commands

1. Display all installed packages

```
(base) C:\Users\AFC>conda list
# packages in environment at E:\ana:
#
# Name                                Version                                Build                                Channel
_anaconda_depends                    2025.12                               py313_mkl_0                         py313h050da96_0
aiobotocore                          2.25.0                               py313h050da96_0                     py313h050da96_0
aiodns                               3.5.0                                py313h050da96_1                     py313h050da96_0
aiohappyeyeballs                    2.6.1                                py313h050da96_0                     py313h050da96_0
aiohttp                              3.13.2                               py313h050da96_0                     py313h050da96_0
aioitertools                        0.12.0                               py313h050da96_0                     py313h050da96_0
aiosignal                           1.4.0                                py313h050da96_0                     py313h050da96_0
alabaster                           0.7.16                               py313h050da96_0                     py313h050da96_0
altair                               5.5.0                                py313h050da96_0                     py313h050da96_0
anaconda-anon-usage                  0.7.5                                pyhb46e38b_100                      py313h050da96_0
anaconda-auth                        0.12.0                               py313h050da96_0                     py313h050da96_0
anaconda-catalogs                   0.2.0                                py313h050da96_3                     py313h050da96_0
anaconda-cli-base                    0.7.0                                py313h050da96_0                     py313h050da96_0
anaconda-client                      1.14.0                               py313h050da96_0                     py313h050da96_0
anaconda-navigator                   2.7.0                                py313h050da96_1                     py313h050da96_0
anaconda-project                     0.11.1                               py313h050da96_1                     py313h050da96_0
anaconda_powershell_prompt          1.1.0                                haa95532_1                          py313h050da96_0
anaconda_prompt                      1.1.0                                haa95532_1                          py313h050da96_0
annotated-types                      0.6.0                                py313h050da96_1                     py313h050da96_0
anyio                                4.10.0                               py313h050da96_0                     py313h050da96_0
aom                                  3.12.1                               h00a0c3c_0                          py313h050da96_0
appdirs                              1.4.4                                pyhd3eb1b0_0                        py313h050da96_0
archspec                             0.2.5                                pyhd3eb1b0_0                        py313h050da96_0
argon2-cffi                          21.3.0                               pyhd3eb1b0_0                        py313h050da96_0
argon2-cffi-bindings                25.1.0                               py313h050da96_0                     py313h050da96_0
arrow                                1.4.0                                py313h050da96_0                     py313h050da96_0
arrow-cpp                            21.0.0                               h6358905_1                          py313h050da96_0
astroid                              3.3.11                               py313h050da96_0                     py313h050da96_0
astropy                              7.1.1                                py313h540bb41_0                     py313h050da96_0
astropy-iers-data                    0.2025.11.10.0.38.31                py313h050da96_0                     py313h050da96_0
```

2. Display a specific package

```
(base) C:\Users\AFC>conda list numpy
# packages in environment at E:\ana:
#
# Name                                Version                                Build                                Channel
numpy                                2.3.5                                py313h050da96_0                     py313h050da96_0
numpy-base                           2.3.5                                py313h1e017a8_0                     py313h050da96_0
numpydoc                              1.9.0                                py313h050da96_0                     py313h050da96_0
```

3. Install flask

```
(base) C:\Users\AFC>conda install flask
3 channel Terms of Service accepted
Retrieving notices: done
Channels:
- defaults
Platform: win-64
Collecting package metadata (repodata.json): done
Solving environment: done

## Package Plan ##

  environment location: E:\ana

added / updated specs:
- flask

The following packages will be downloaded:
```

package	build	
conda-26.1.0	py313haa95532_0	1.2 MB
openssl-3.0.19	hbb43b14_0	6.9 MB
Total:		8.2 MB

```
The following packages will be UPDATED:

ca-certificates      2025.11.4-haa95532_0 --> 2025.12.2-haa95532_0
certifi              2025.11.12-py313haa95532_0 --> 2026.01.04-py313haa95532_0
conda                25.11.0-py313haa95532_0 --> 26.1.0-py313haa95532_0
openssl              3.0.18-h543e019_0 --> 3.0.19-hbb43b14_0
```

```
(base) C:\Users\AFC>conda list flask
# packages in environment at E:\ana:
#
# Name                      Version      Build                      Channel
flask                       3.1.2       py313haa95532_0
```

4. Update any package

```
(base) C:\Users\AFC>conda update numpy
3 channel Terms of Service accepted
Channels:
- defaults
Platform: win-64
Collecting package metadata (repodata.json): done
Solving environment: done

# All requested packages already installed.
```

5. Remove a package

```
(base) C:\Users\AFC>conda remove flask
3 channel Terms of Service accepted
Channels:
- defaults
Platform: win-64
Collecting package metadata (repodata.json): done
Solving environment: done

## Package Plan ##

  environment location: E:\ana

removed specs:
- flask

The following packages will be REMOVED:

_anaconda_depends-2025.12-py313_mkl_0
aiobotocore-2.25.0-py313haa95532_0
aioitertools-0.12.0-py313haa95532_0
alabaster-0.7.16-py313haa95532_0
altair-5.5.0-py313haa95532_0
anyio-4.10.0-py313haa95532_0
appdirs-1.4.4-pyhd3eb1b0_0
argon2-cffi-21.3.0-pyhd3eb1b0_0
argon2-cffi-bindings-25.1.0-py313h02ab6af_0
arrow-1.4.0-py313haa95532_0
arrow-cpp-21.0.0-h6358905_1
astroid-3.3.11-py313haa95532_0
astropy-7.1.1-py313h540bb41_0
astropy-iers-data-0.2025.11.10.0.38.31-py313haa95532_0
```

```
Proceed ([y]/n)? y
```

```
Downloading and Extracting Packages:
```

```
Preparing transaction: done
Verifying transaction: done
Executing transaction: done
```

Task 2: Virtual Environment Commands

1. Create environment

```
(base) C:\Users\AFC>conda create --name Izaz_B22F0001AI029 python=3.10
```

```
3 channel Terms of Service accepted
```

```
Channels:
```

```
- defaults
```

```
Platform: win-64
```

```
Collecting package metadata (repodata.json): done
```

```
Solving environment: done
```

```
## Package Plan ##
```

```
environment location: E:\ana\envs\Izaz_B22F0001AI029
```

```
added / updated specs:
```

```
- python=3.10
```

```
The following packages will be downloaded:
```

package	build	
expat-2.7.4	hd7fb8db_0	32 KB
libexpat-2.7.4	hd7fb8db_0	121 KB
packaging-25.0	py310haa95532_1	165 KB
pip-26.0.1	pyhc872135_0	1.1 MB
python-3.10.19	h981015d_0	15.3 MB
setuptools-80.10.2	py310haa95532_0	1.3 MB
sqlite-3.51.1	hda9a48d_0	917 KB
tzdata-2025c	he532380_0	118 KB
vc-14.3	h2df5915_10	19 KB
wheel-0.46.3	py310haa95532_0	78 KB
Total:		19.2 MB

```
The following NEW packages will be INSTALLED:
```

```
Proceed ([y]/n)? y
```

```
Downloading and Extracting Packages:
```

```
Preparing transaction: done
```

```
Verifying transaction: done
```

```
Executing transaction: done
```

```
#
```

```
# To activate this environment, use
```

```
#
```

```
#     $ conda activate Izaz_B22F0001AI029
```

```
#
```

```
# To deactivate an active environment, use
```

```
#
```

```
#     $ conda deactivate
```

2. Activate environment

```
(base) C:\Users\AFC>conda activate Izaz_B22F0001AI029
```

```
(Izaz_B22F0001AI029) C:\Users\AFC>|
```

3. List environments

```
(Izaz_B22F0001AI029) C:\Users\AFC>conda env list
```

```
# conda environments:
```

```
#
```

```
# * -> active
```

```
# + -> frozen
```

```
base                                E:\ana
```

```
Izaz_B22F0001AI029      *         E:\ana\envs\Izaz_B22F0001AI029
```

4. Install Jupyter inside environment

```
(Izaz_B22F0001AI029) C:\Users\AFC>conda install jupyter
```

```
3 channel Terms of Service accepted
```

```
Channels:
```

```
- defaults
```

```
Platform: win-64
```

```
Collecting package metadata (repodata.json): done
```

```
Solving environment: done
```

```
## Package Plan ##
```

```
environment location: E:\ana\envs\Izaz_B22F0001AI029
```

```
added / updated specs:
```

```
- jupyter
```

```
The following packages will be downloaded:
```

package	build	
anyio-4.10.0	py310h02ab6af_0	230 KB
argon2-cffi-25.1.0	py310h02ab6af_0	31 KB
argon2-cffi-bindings-25.1.0	py310h02ab6af_0	41 KB
asttokens-3.0.0	py310h02ab6af_0	41 KB
async-lru-2.0.5	py310h02ab6af_0	18 KB
attrs-25.4.0	py310h02ab6af_0	152 KB
babel-2.17.0	py310h02ab6af_0	13.3 MB
beautifulsoup4-4.14.3	py310h02ab6af_0	212 KB
bleach-6.3.0	py310h02ab6af_0	84 KB
brotlicffi-1.2.0.0	py310h02ab6af_0	346 KB
certifi-2026.01.04	py310h02ab6af_0	148 KB
cffi-2.0.0	py310h02ab6af_1	241 KB

5. Add environment to Jupyter kernel

```
(Izaz_B22F0001AI029) C:\Users\AFC>python -m ipykernel install --user --name Izaz_B22F0001AI029
```

```
Installed kernelspec Izaz_B22F0001AI029 in C:\Users\AFC\AppData\Roaming\jupyter\kernels\izaz_b22f0001ai029
```

6. Deactivate environment

```
(Izaz_B22F0001AI029) C:\Users\AFC>conda deactivate
```

```
(base) C:\Users\AFC>
```

7. Remove environment

```
(base) C:\Users\AFC>conda remove --name Izaz_B22F0001AI029 --all
3 channel Terms of Service accepted

Remove all packages in environment E:\ana\envs\Izaz_B22F0001AI029:

## Package Plan ##

  environment location: E:\ana\envs\Izaz_B22F0001AI029

The following packages will be REMOVED:

anyio-4.10.0-py310haa95532_0
argon2-cffi-25.1.0-py310haa95532_0
argon2-cffi-bindings-25.1.0-py310h02ab6af_0
asttokens-3.0.0-py310haa95532_0
async-lru-2.0.5-py310haa95532_0
attrs-25.4.0-py310haa95532_2
```

Task 3: Create Environment in Anaconda Navigator (GUI)

Create new environment

Name:

Izazkhan_B23F0001AI029

Location:

E:\ana\envs\Izazkhan_B23F0001AI029

Packages:

☒ Python

3.13.11

▼

☐ R

3.6.1

▼

Cancel

Create

Summary

In this lab, I practiced basic Conda commands and learned how to manage Python packages and virtual environments using Anaconda. I displayed installed packages, installed and updated libraries, and removed packages to understand environment management. I also created a virtual environment, activated it, installed Jupyter inside it, and added it to the Jupyter kernel list. Finally, I used Anaconda Navigator to create an environment through the graphical interface. This lab improved my understanding of package control, environment isolation, and workflow management in Python.